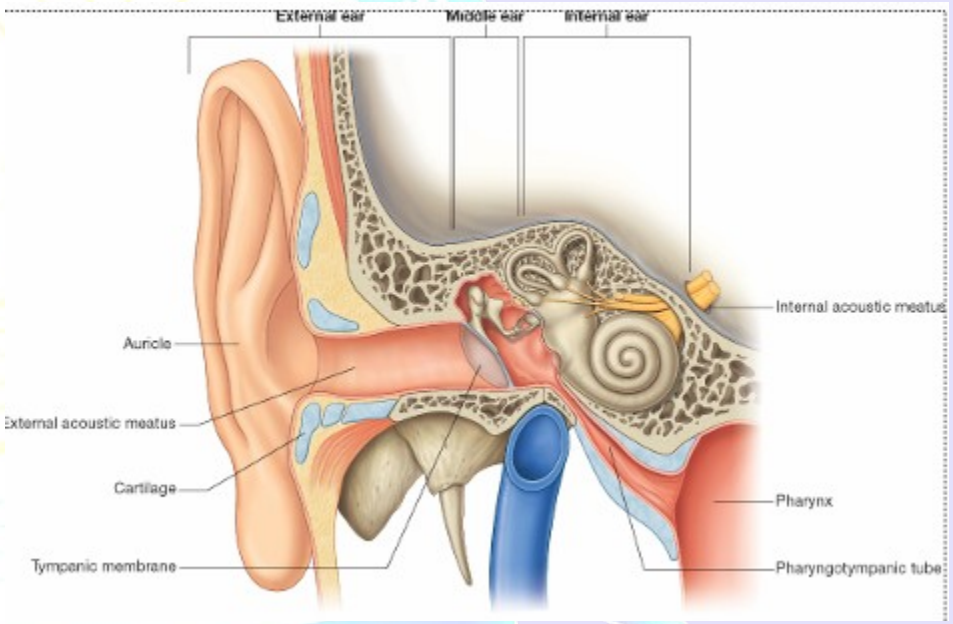
INTERNAL EAR

Learning objective

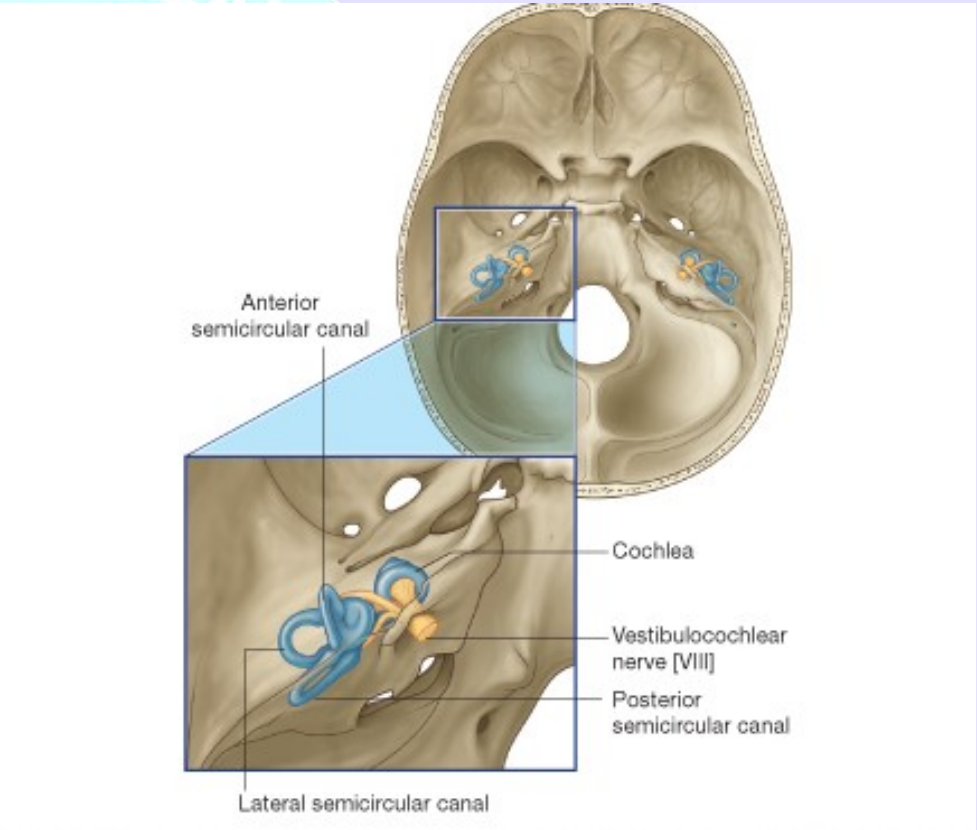
- At the end of the lecture the student should be able to know:
- Details of internal ear.
- Parts of internal ear.
- Functions of internal ear.
- Details of organ of hearing and balance.
- Applied anatomy

Ear

- The ear is the organ of hearing and balance. It has three parts:
- **External ear**
- **Middle ear**
- **Internal ear**

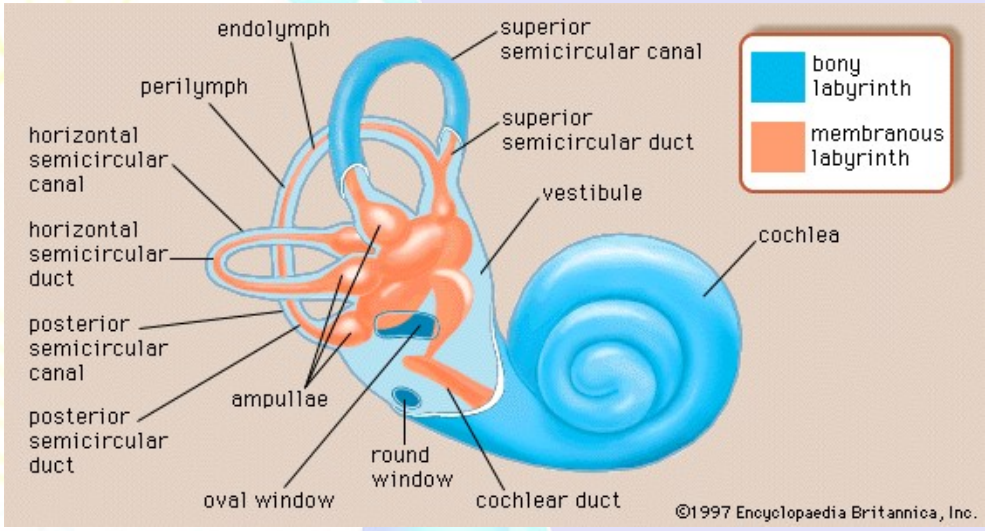


- All these structures are in the petrous part of the temporal bone between the middle ear laterally and the internal acoustic meatus medially.



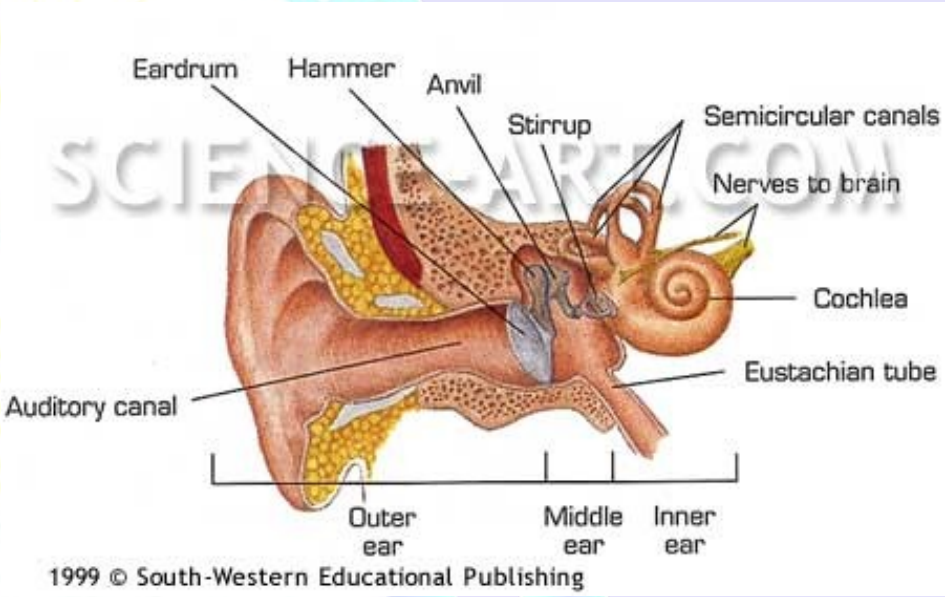
Parts of internal ear

- Internal ear consists of:
- **Bony labyrinth**
  - Vestibule,
  - Three semicircular canals
  - Cochlea
- **Membranous labyrinth.**
  - Semicircular ducts
  - Cochlear duct
  - Two sacs (utricle and saccule).
- These bony cavities are lined with periosteum and contain a clear fluid the **perilymph**.
- The membranous spaces are filled with **endolymph**.



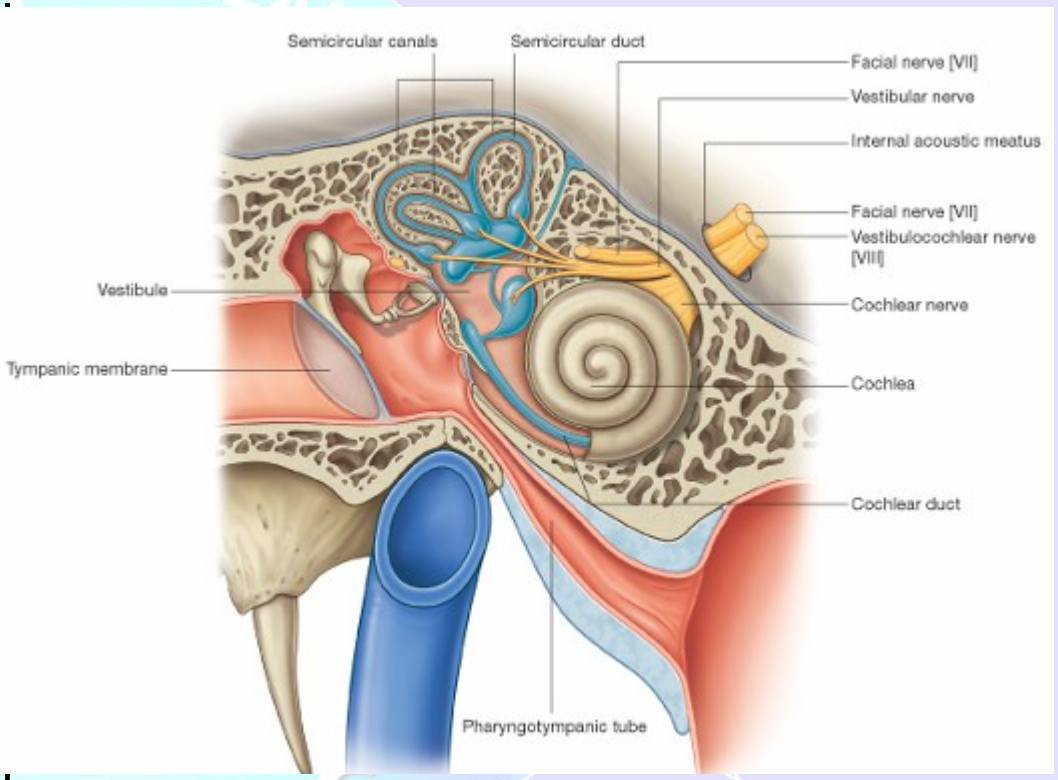
Function of internal ear

- The internal ear converts the mechanical signals received from the middle ear, which start as sound captured by the external ear, into electrical signals to transfer information to the brain.
- The internal ear also contains receptors that detect motion and position.



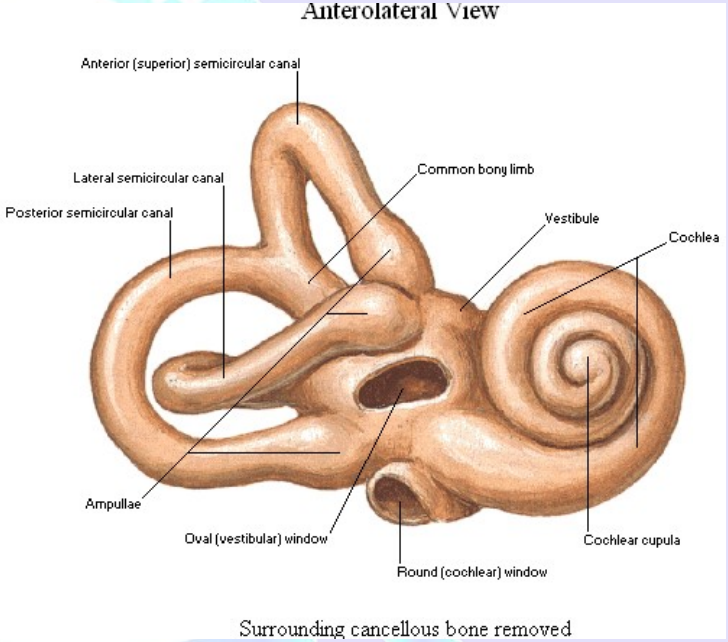
INTERNAL EAR

- The structures in the interna ear convey information to th brain about balance and hearing:
- Cochlear duct is the organ c hearing
- Semicircular ducts, utricle, a saccule are the organs of balance.
- The nerve responsible for these functions is the vestibulocochlear nerve [VII which divides into vestibular (balance) and cochlear (hearing) parts after entering the internal acoustic meatus



Bony labyrinth

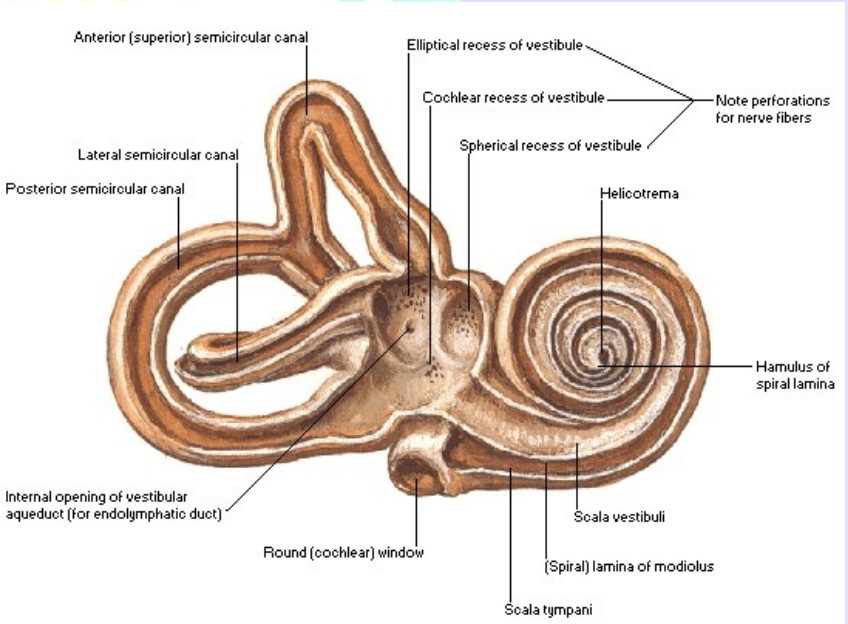
- **<u>Vestibule</u>**,
- It is the central part of the bony labyrinth.
- contains the oval window in its lateral wall
- It communicates anteriorly with the cochlea and posterosuperiorly with the semicircular canals.
- A narrow canal (the **vestibular aqueduct**) leaves the vestibule, and passes through the temporal bone to open on the posterior surface of the petrous part of the temporal bone.



Bony labyrinth

Semicircular canals

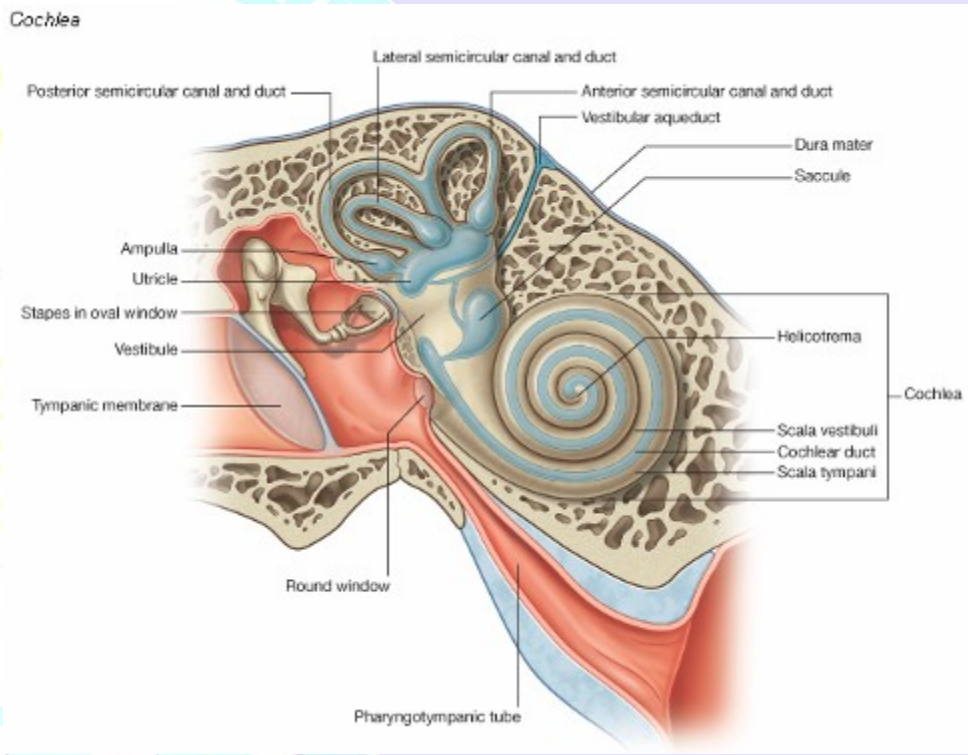
- Projecting in a posterosuperior direction from the vestibule are three in nummber
  - **Anterior (superior) vertical**
  - **Posterior (vertical)**
  - **Lateral (horizontal)**
- Each of these canals forms two-thirds of a circle connected at both ends to the vestibule and with one end dilated to form the **ampulla**.
- The canals are oriented so that each canal is at right angles to the other two.



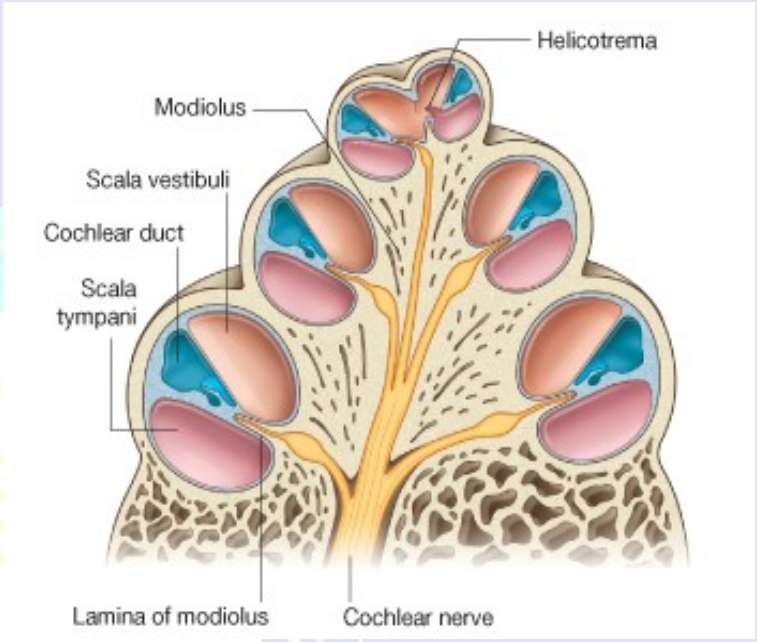
Bony labyrinth

Cochlea

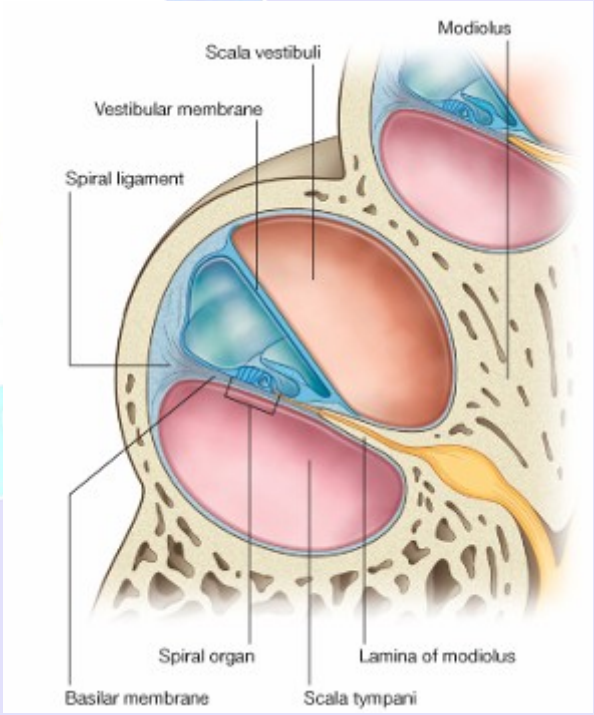
- It is a bony structure that twists on itself two and one-half to two and three-quarter times around a central column of bone (the **modiolus**).
- This arrangement produces a cone-shaped structure with a **base of cochlea** that faces posteromedially and an apex that faces anterolaterally.
- This positions the wide base of the modiolus near the internal acoustic meatus, where it is entered by branches of the cochlear part of the vestibulocochlear nerve [VIII].



Bony labyrinth

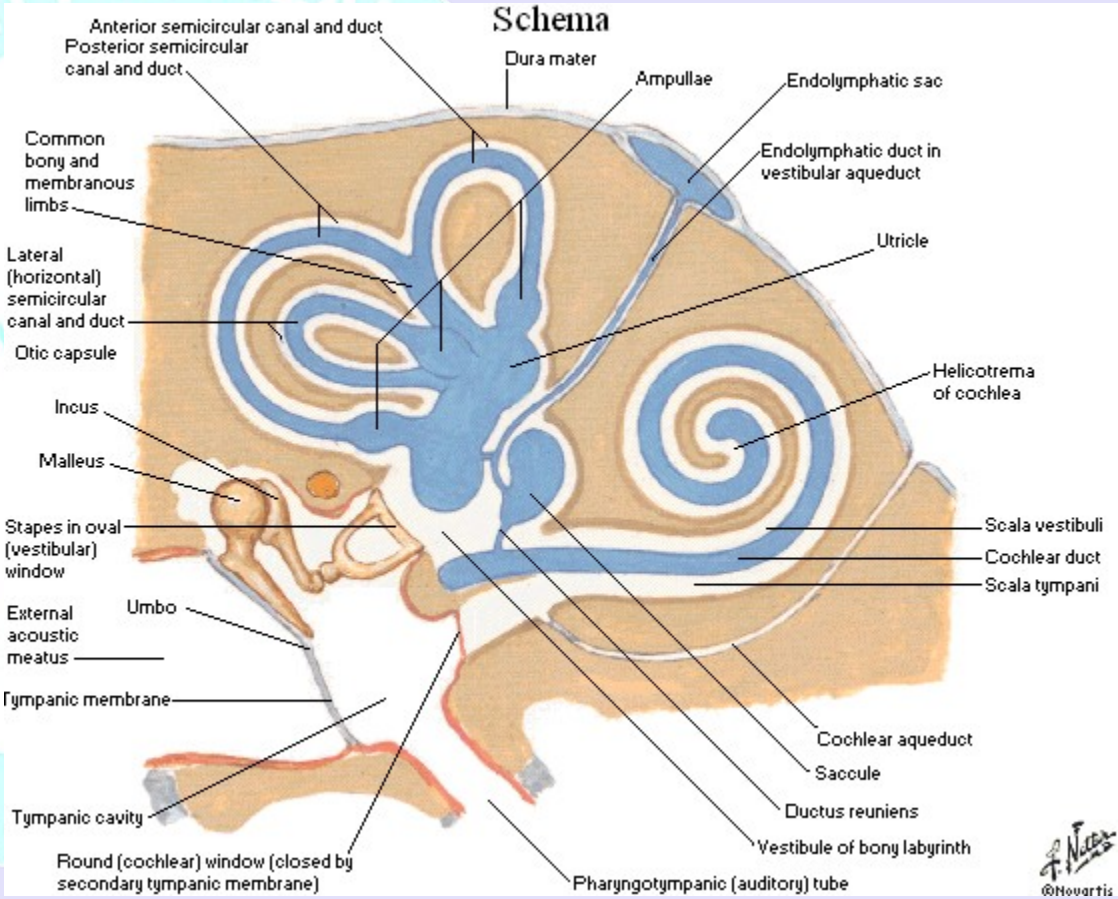


- **Lamina of modiolus**, or **spiral lamina**).
- Circling around the modiolus, and held in a central position by its attachment to the lamina of modiolus, is the cochlear duct, which is a component of the membranous labyrinth.
- Cochlear duct creates 2 canals
  - **Scala vestibuli**
  - Continuous with the vestibule
  - **Scala tympani**
  - Separated from the middle ear by the secondary tympanic membrane covering the round window
- They extend throughout the cochlea and are continuous with each other at the apex through a narrow slit (the **helicotrema**)



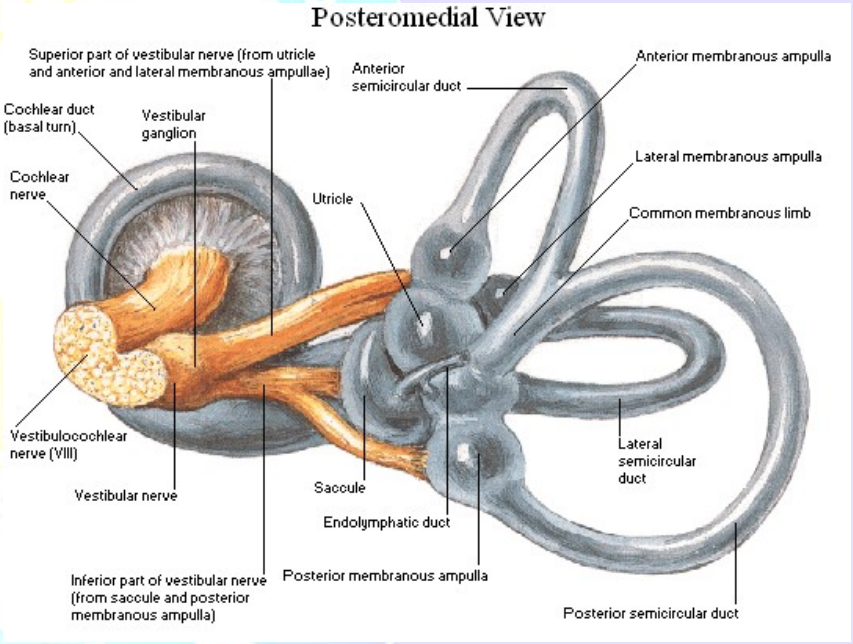
Cochlear aqueduct

- Near the round window is a small channel (the **cochlear canaliculus**), which passes through the temporal bone and opens on its inferior surface into the posterior cranial fossa.
- This provides a connection between the perilymph-containing cochlea and the subarachnoid space.



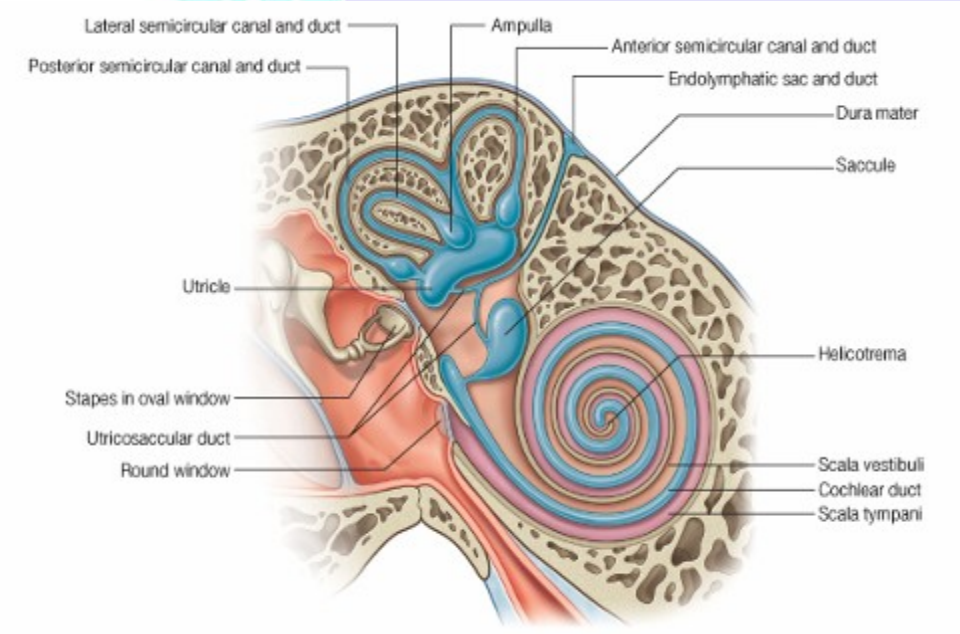
Membranous labyrinth

- It is a continuous system of ducts and sacs within the bony labyrinth.
- It is filled with endolymph and separated from the periosteum that covers the walls of the bony labyrinth by perilymph.
- Consisting of
- Two sacs (the utricle and the saccule) and
- Four ducts (the three semicircular ducts and the cochlear duct).
- The general organization of the parts of the membranous labyrinth places:
- Cochlear duct within the cochlea of the bony labyrinth, anteriorly;
- Three semicircular ducts within the three semicircular canals of the bony labyrinth, posteriorly;
- Saccule and utricle within the vestibule of the bony labyrinth, in the middle.



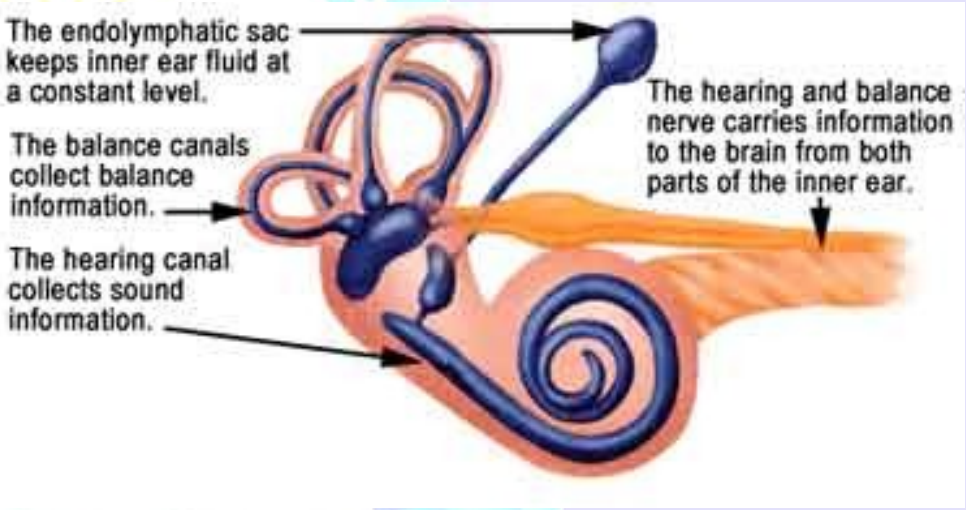
ORGANS OF BALANCE

- 5 of the 6 components of the membranous labyrinth are concerned with balance. These are the:
- Two sacs (the utricle and the saccule)
- Three ducts (anterior, posterior, and lateral semicircular ducts).
- Utricle
- The utricle is the larger of the two sacs. It is oval, elongated and irregular in shape and is in the posterosuperior part of the vestibule of the bony labyrinth.
- Three semicircular ducts empty into the utricle
- Saccule
- The saccule is a smaller, rounded sac lying in the anteroinferior part of the vestibule of the bony labyrinth. The cochlear duct empties into it.
- The **utriculosaccular duct** establishes continuity between all components of the membranous labyrinth and connects the utricle and saccule



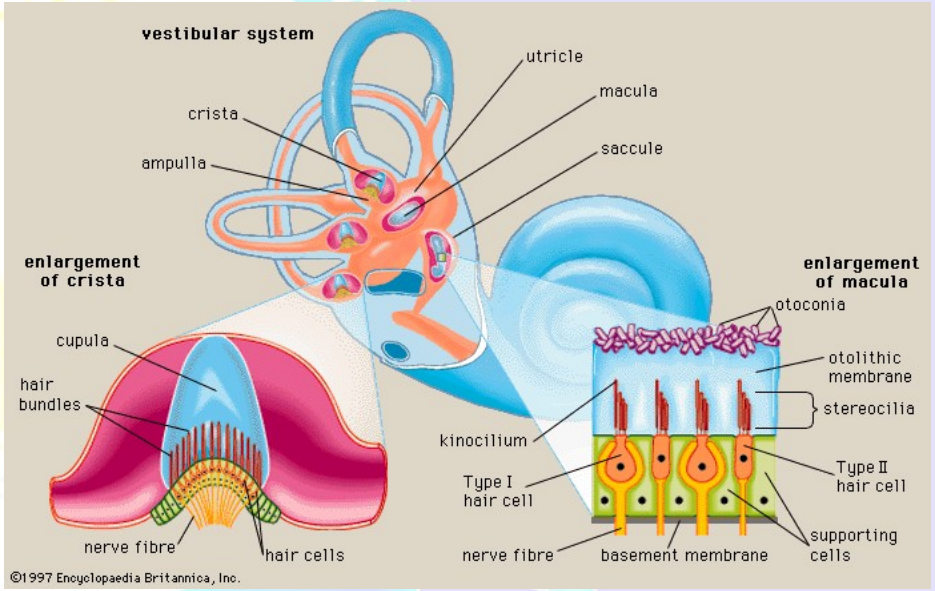
ORGANS OF BALANCE

- Branching from utricosaccular duct is the **endolymphatic duct**, which enters the vestibular aqueduct (a channel through the temporal bone) to emerge onto the posterior surface of the petrous part of the temporal bone in the posterior cranial fossa.
- Here the endolymphatic duct expands into the **endolymphatic sac**, which is an extradural pouch.
- Each semicircular duct is similar in shape, including a dilated end forming the ampulla, to its complementary bony semicircular canal, only much smaller.



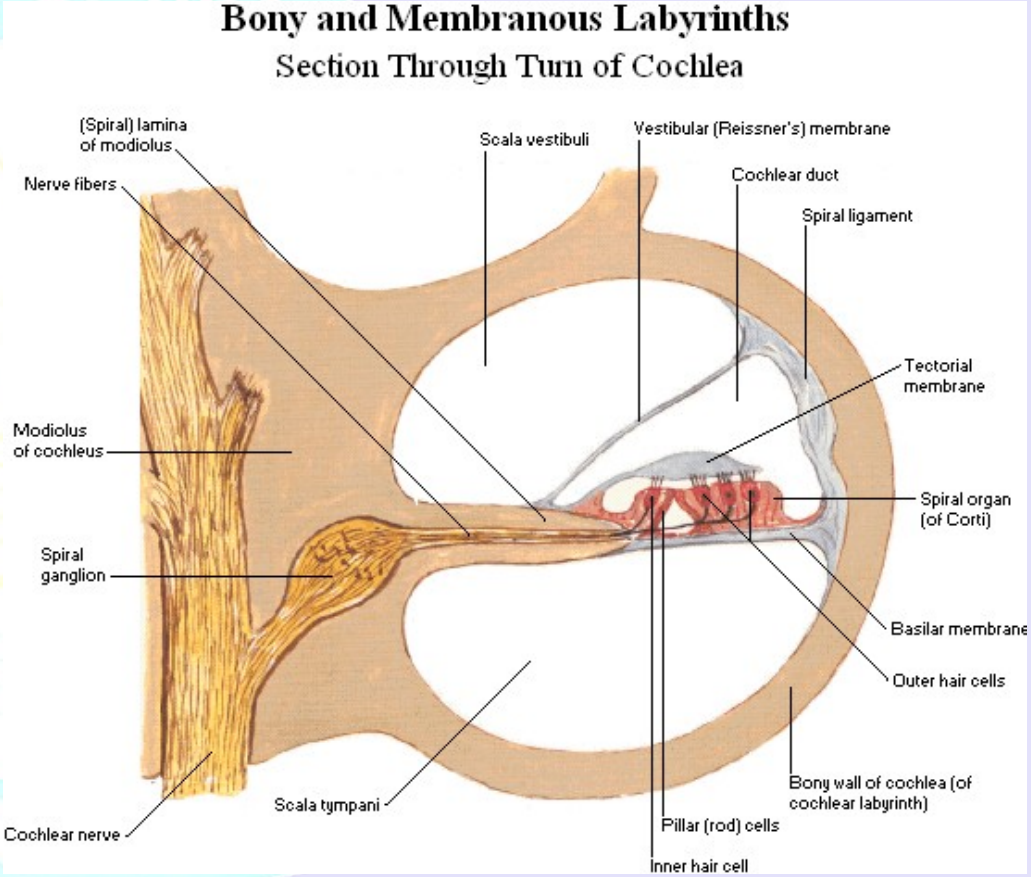
SENSORY RECEPTORS

- Functionally, sensory receptors for balance are organized into unique structures and in each of the components of the vestibular apparatus.
- In the utricle and saccule this sense organ is the **macula of utricle** and **macula of saccule**,
- Ampulla of each of the three semicircular ducts it is the **crista.**
- Utricle responds to centrifugal and vertical acceleration,
- Saccule responds to linear acceleration.
- Receptors in the three semicircular ducts respond to movement in any direction.



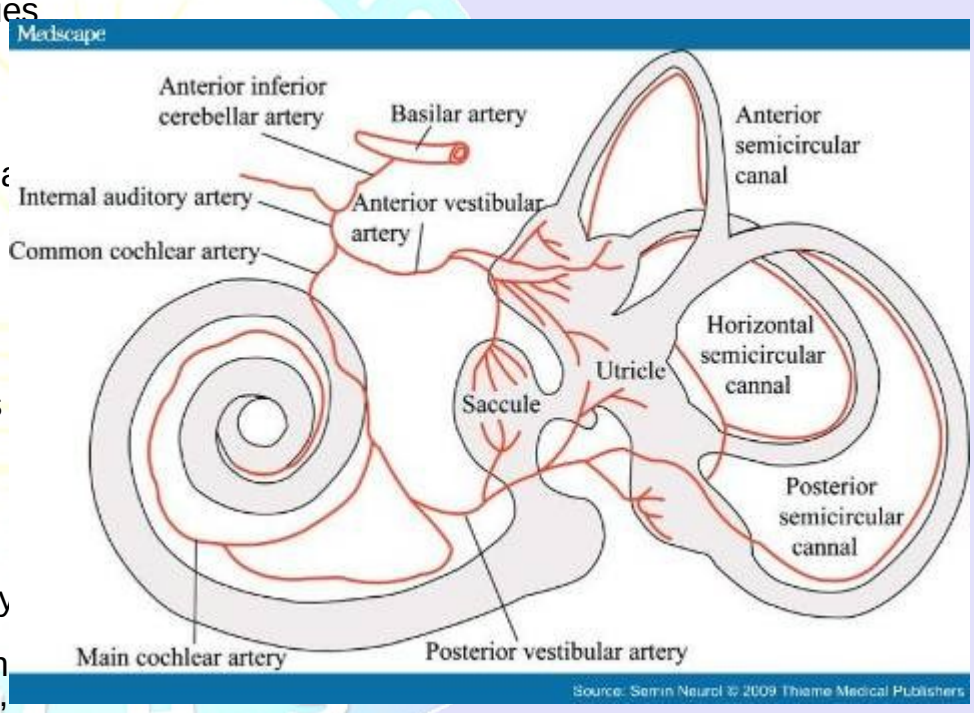
ORGAN OF HEARING

- Triangular-shaped cochlear duct has:
- An outer wall against the bony cochlea consisting of thickened, epithelial-lined periosteum (the **spiral ligament**);
- **Roof (vestibular surface membrane)** which separates the endolymph in the cochlear duct from the perilymph in the scala vestibuli and consists of a membrane with a connective tissue core lined on either side with epithelium;
- **Floor**, which separates the endolymph in the cochlear duct from the perilymph in the scala tympani and consists of the free edge of the lamina of modiolus, and a membrane (**basilar membrane**) extending from this free edge of the lamina of modiolus to an extension of the spiral ligament covering the outer wall of the cochlea.
- **Spiral organ** is the organ of hearing, rests on the basilar membrane, and projects into the enclosed, endolymph-filled cochlear duct.



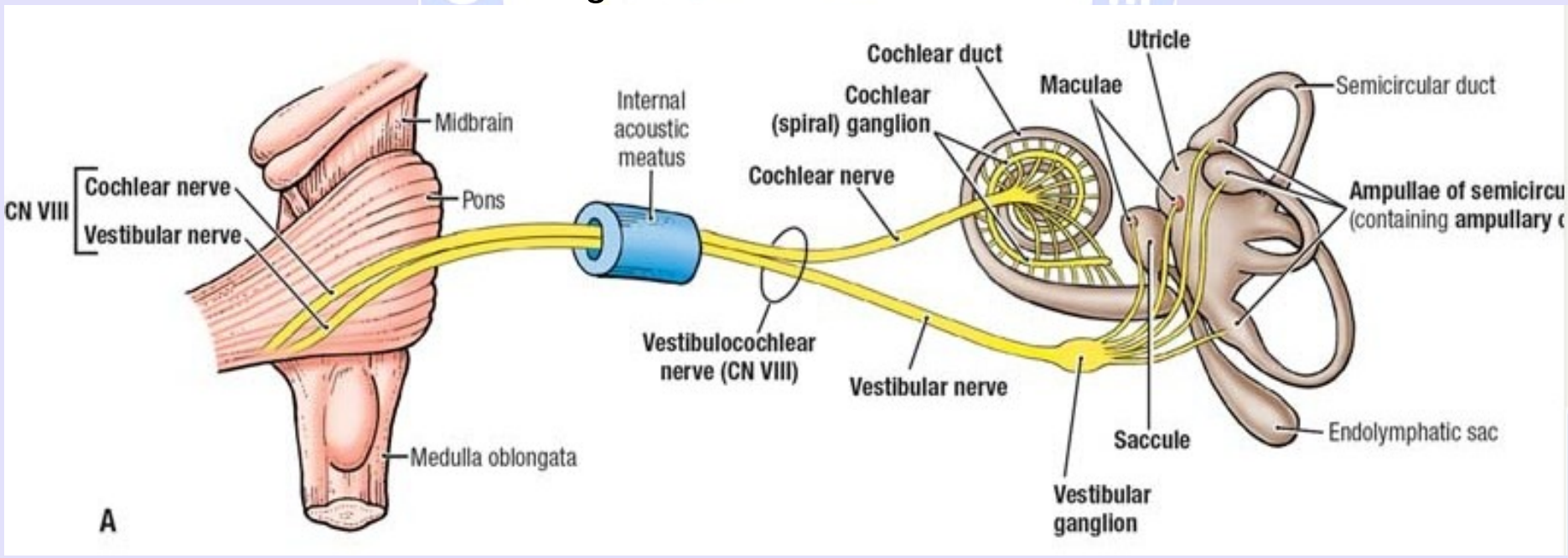
Vessels

- Arterial supply to the internal ear is divided between vessels supplying the bony labyrinth and the membranous labyrinth.
- **Bony labyrinth** is supplied by the same arteries that supply the surrounding temporal bone- these include
- Anterior tympanic branch from the maxillary artery
- stylomastoid branch from the posterior auricula artery
- Petrosal branch from the middle meningeal artery.
- **Membranous labyrinth**
- **labyrinthine artery**, which either arises from the anterior inferior cerebellar artery or is a direct branch of the basilar artery-whatever its origin, it enters the internal acoustic meatus with the facial [VII] and glossopharyngeal [IX] nerves and eventually divides into:
- **cochlear branch**, which passes through the modiolus and supplies the cochlear duct;
- one or two **vestibular branches**, which supply the vestibular apparatus.
- Venous drainage of the membranous labyrinth is through vestibular veins and cochlear veins, which follow the arteries. These come together to form a **labyrinthine vein**, which eventually empties into either the inferior petrosal sinus or the sigmoid sinus.



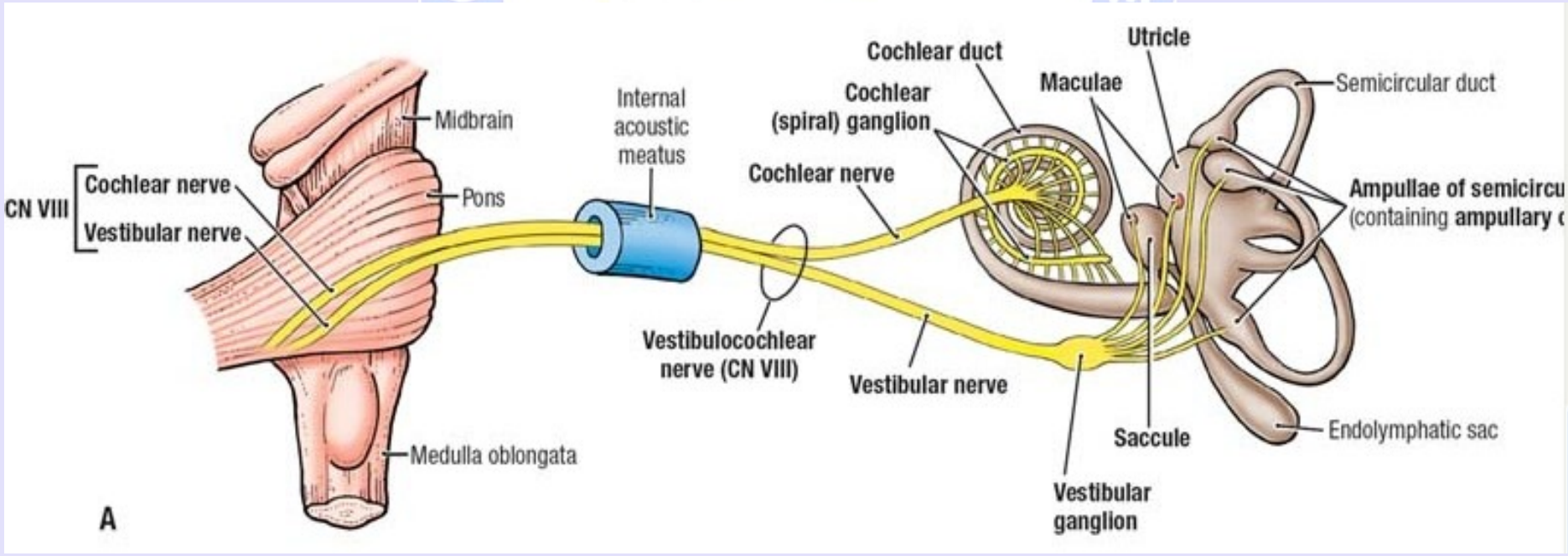
INNERVATION

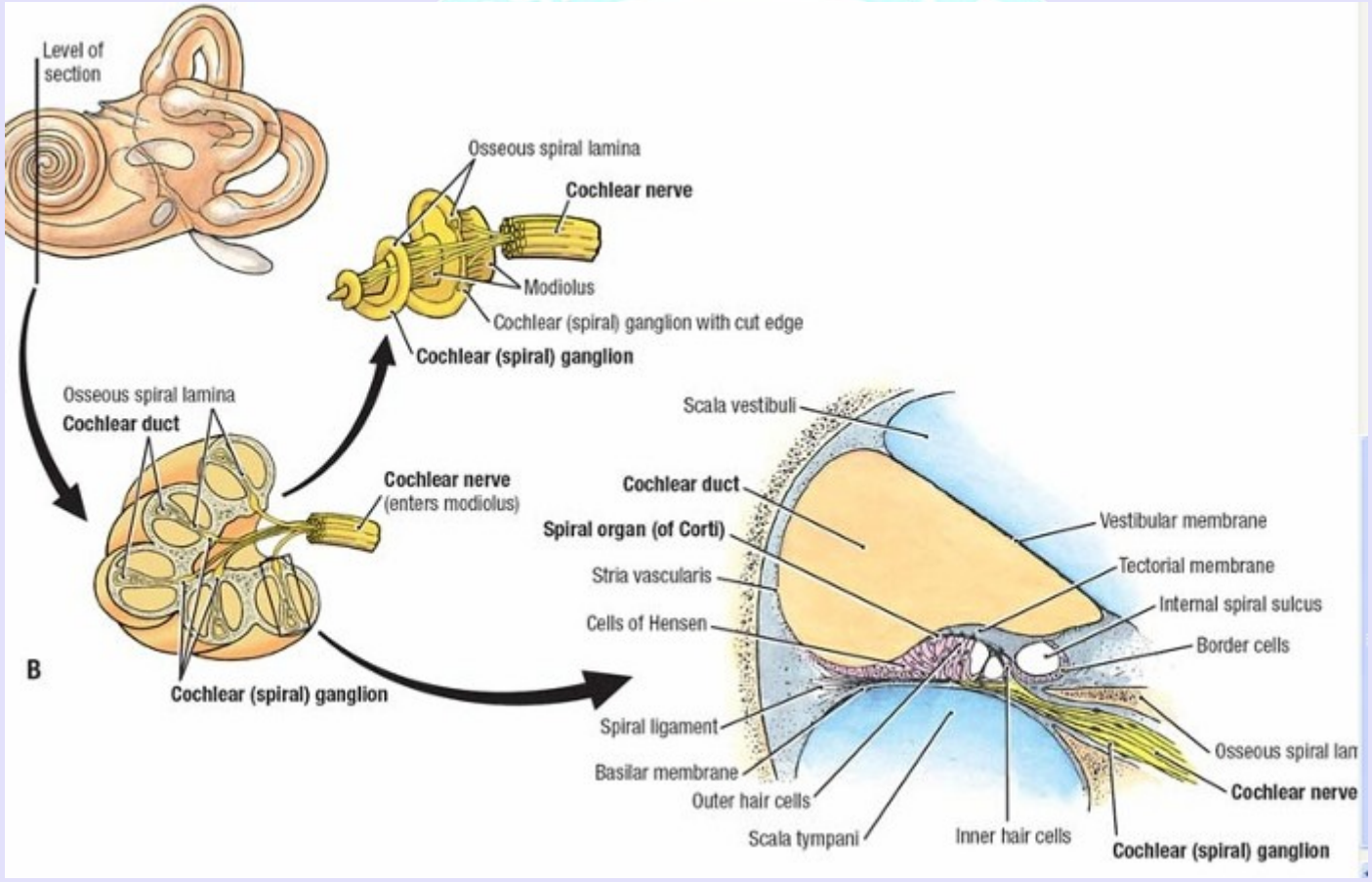
- The vestibulocochlear nerve [VIII] carries special afferent fibers for hearing (the cochlear component) and balance (the vestibular component).
- It enters the lateral surface of the brainstem, between the pons and medulla, after exiting the



INNERVATION

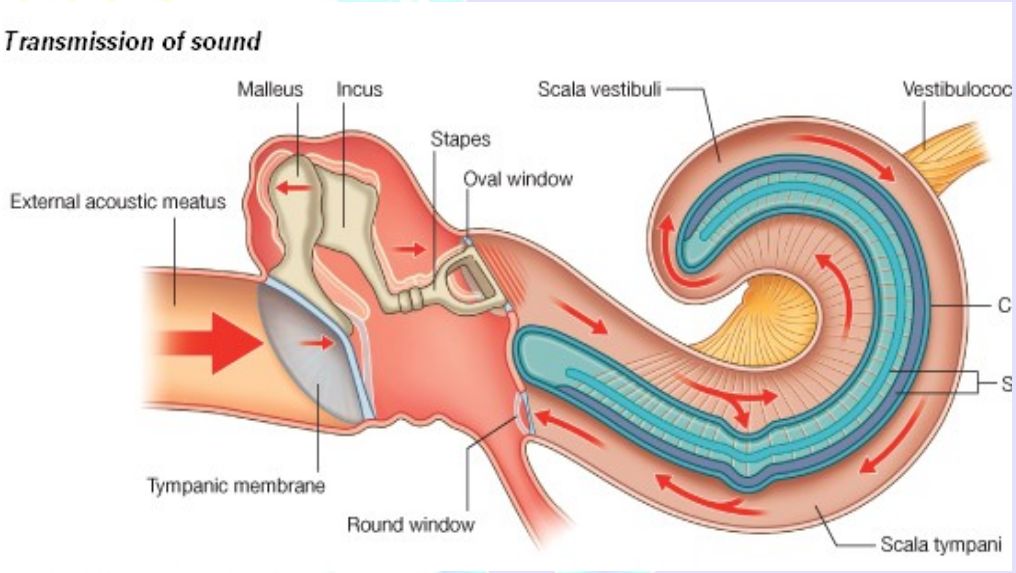
- The vestibular nerve enlarges to form the **vestibular ganglion**, before dividing into **superior** and **inferior parts**, which distribute to the three semicircular ducts and the utricle and saccule.
- The cochlear nerve enters the base of the cochlea and passes upwards




Level of section
Osseous spiral lamina
Cochlear nerve
Modiolus
Cochlear (spiral) ganglion with cut edge
Cochlear (spiral) ganglion
Osseous spiral lamina
Cochlear duct
Cochlear nerve
(enters modiolus)
B
Cochlear (spiral) ganglion
Scala vestibuli
Cochlear duct
Spiral organ (of Corti)
Stria vascularis
Cells of Hensen
Spiral ligament
Basilar membrane
Outer hair cells
Scala tympani
Inner hair cells
Vestibular membrane
Tectorial membrane
Internal spiral sulcus
Border cells
Osseous spiral lan
Cochlear nerve
Cochlear (spiral) ganglion

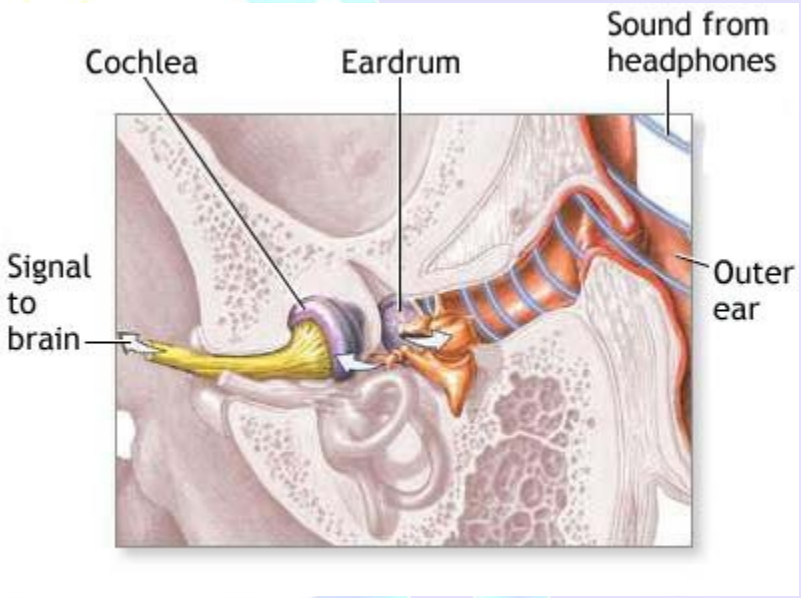
TRANSMISSION OF SOUND

- A sound wave enters the external acoustic meatus and strikes the tympanic membrane moving it medially.
- As the handle of malleus is attached to this membrane the handle of malleus also moves medially.
- This moves the head of the malleus laterally. Because the heads of the malleus and incus articulate with each other, the head of the incus is also moved laterally.
- This pushes the long process of the incus medially. The long process articulates with the stapes, so its movement causes the stapes to move medially.
- In turn, because the base of stapes is attached to the oval window, the oval window is also moved medially.
- This action completes the transfer of a large-amplitude, low-force, airborne wave that vibrates the tympanic membrane into a small-amplitude, high-force vibration of the oval window, which generates a wave in the fluid-filled scala vestibuli of the cochlea.

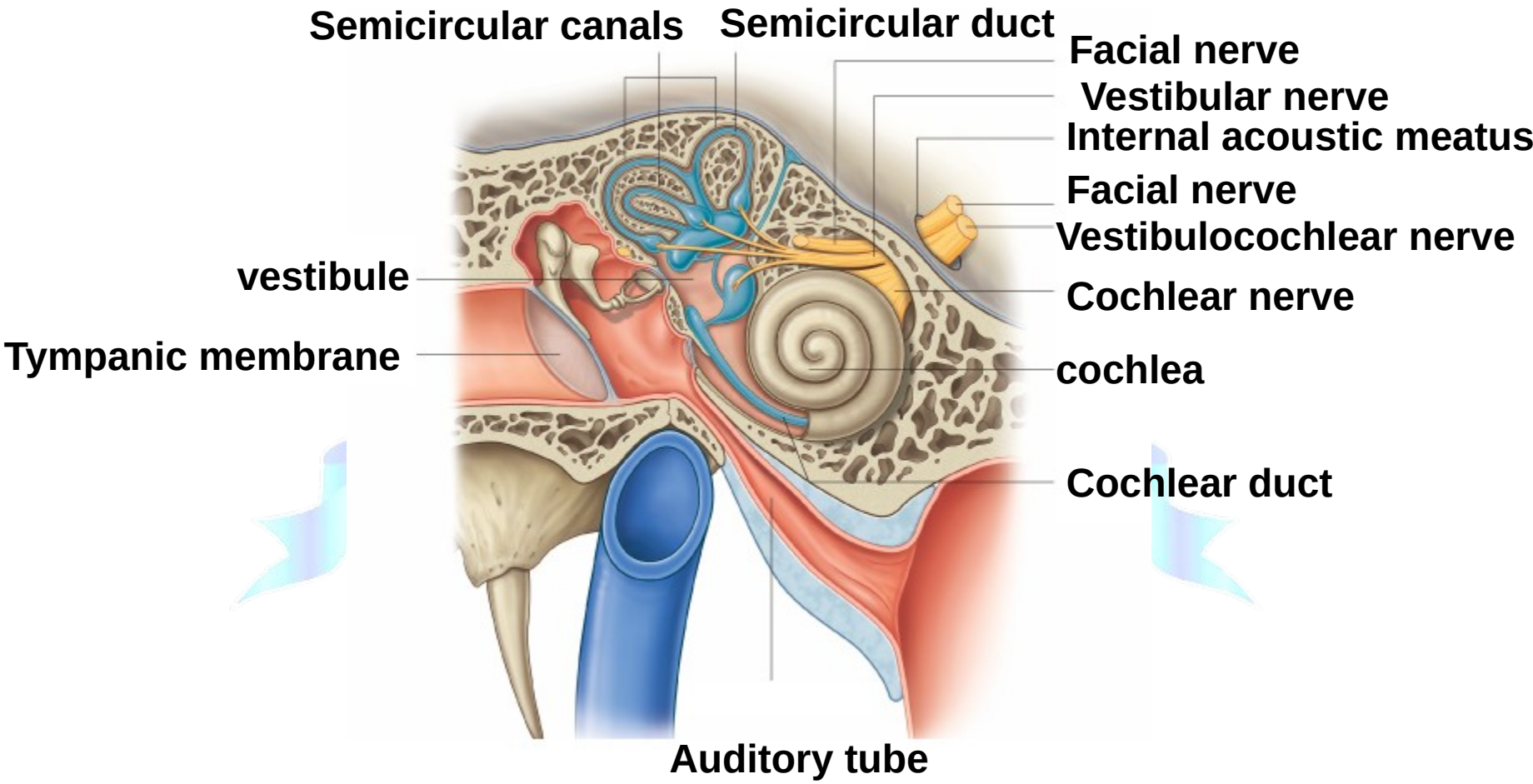


TRANSMISSION OF SOUND

- The wave established in the perilymph of the scala vestibuli moves through the cochlea and causes an outward bulging of the secondary tympanic membrane covering the round window at the lower end of the scala tympani.
- This causes the basilar membrane to vibrate, which in turn leads to stimulation of receptor cells in the spiral organ.
- The receptor cells send impulses back to the brain through the cochlear part of the vestibulocochlear nerve [VIII] where they are interpreted as sound.
- If the sounds are too loud, causing excessive movement of the tympanic membrane, contraction of the tensor tympani muscle (attached to the malleus) and/or the stapedius muscle (attached to the stapes) dampens the vibrations of the ossicles and decreases the force of the vibrations reaching the oval window.



Identify?


Semicircular canals
Semicircular duct
Facial nerve
Vestibular nerve
Internal acoustic meatus
Facial nerve
Vestibulocochlear nerve
vestibule
Cochlear nerve
Tympanic membrane
cochlea
Cochlear duct
Auditory tube



thanks